Seminar zur Algebraischen Geometry program :
A path towards semistable models of smooth projective curves over $\mathbb{Q}$

Important informations :

This seminar will be hold during the winter semester 2026/27, on Tuesdays, between 2 :00 - 4 :00 p.m. in SR 1C. Careful that my French habits will make the seminar start at 2 :00 p.m. sharp, and end somewhere between 3 :30 and 4 :00 p.m. depending on the week.

Warning : Nov 24th is replaced by Nov 27th, 10 :00 - 12 :00 a.m. in SRZ 204.

You should register online to the lesson if you want to validate it. Here is the link : `https://www.uni-muenster.de/LearnWeb/learnweb2/course/view.php?id=95213`. We can have a meeting so that I can give details about what I expect from the talk, help you naviguate the references or solve some issues. Ideally, this meeting would take place on Tuesday afternoon after the previous talk, or the following wednesday morning.

Do not hesitate to contact me by email for scheduling the appointment or for questions. My email is n.marquis@uni-muenster.de and you can also find all informations on my webpage `https://nmarquis-maths.github.io/teaching.html`.

Finally, if you don't find the book, or .pdf of some reference, please ask me. You can also discuss with me if you think you need to adapt the length to be able to explain properly rather than hastily, so that we can maybe make a selection together of what's the most important. The things labelled "if time allows" should only be covered if the program reveals itself too short, and I expect this will not happen.

Talk 1. (Oct. 13, Lilly Steinbauer) Journey to the definition of Dedekind schemes

Aim : we will walk on an avenue of definitions and recalls to reach the definitions of Dedekind schemes of dimension 1 *as well as regular schemes.*

The aim of this first talk is to recall and/or define some very useful properties of rings and schemes that we will be using in the whole seminar. As some may have already seen some of the notions, try to focus on giving formal definitions and intuition via examples rather than running all proofs. I only require the proof of [Sta23, Tag 0357] or of [Sta23, Tag 01ON] using [Sta23, Tag 01OM] to be done (but feel free to add other if you have time and think it is useful). I give mostly *Stacks Project* as reference : it is obviously too thick, hence try to extract what seems the essential for each notion.

We define the notions of integral ([Sta23, Tag 01OJ]) and normal schemes ([Sta23, Tag 037B] for ring theoretic definitions, [Sta23, Tag 033H]) and a few of their characterisation. We then define the dimension of a scheme ([Sta23, Tag 00KD]). You can state just enough properties to then run the examples of $\mathbb{A}^n_k$ say for an algebraically closed field k and for $n \geq 2$, of the "cross" $\mathrm{Spec}(k[x,y]/xy)$ and of $\mathrm{Spec}(\mathbb{Z})$. Define Dedekind schemes (nothing more than [LE02, §4.1, Def 1.2]) and give examples. Then define regular schemes (select among [Sta23, Tag 065U] [Sta23, Tag 02IR]). End by explaining why a normal curve over a field is regular (the key point is [Mat80, Th. 11.2]).

Only if time allows, try to understand why [Sta23, Tag 0B2I] roughly says that a two dimensional irreducible scheme over a Dedekind scheme of dimension 1 can be thought as a family of curves. This can be inserted before defining regular schemes.

Talk 2. (Oct. 20, Sevag Buyuksimkesyan) Projective geometry and smoothness

Aim : we will define and give intuition about projective and smooth schemes/maps.

As projective schemes have been covered in previous classes (though I think it is useful to clear the minds about it) and smoothness is a wide subject, I expect this talk to give precise definitions and intuition rather than proofs if one has to choose. I only require the proof of [LE02, §4.1.1, Lem. 1.18] to be done (but feel free to add other if you have time/ think it is useful). The reference I give are obviously too thick, hence try to extract the essence and useful properties for each notion. You may use the full two hours.

We first define projective scheme associated to a graded ring ([Duca, §6.1] is the clearest reference in my mind, but as it is in French, feel free to find your own). Make a quick analysis of $\mathbb{P}^N_k$, its closed subschemes and of the invertible sheaves $\mathcal{O}_{\mathbb{P}^N_k}(n)$ (see [Duca, §6.2.1-6.2.4, §6.3.10] or [Sta23, Tag 01ND] [Sta23, Tag 01MM]). We then define projective morphisms ([Sta23, Tag 01W7]), which allows to define an arithmetic surface (only [LE02, §8.3.1, Def. 3.1 and 3.14]). Please mention the criterion [LE02, §4.1.1, Lem. 1.18] for being normal.

The last task is to define the notion of a smooth map. I think it is disturbing as a first approach to give the definition using the naive cotangent complex. Rather combine [Sta23, Tag 00T6] [Sta23, Tag 01V7] [Sta23, Tag 01V8] to give a definition that looks more like a jacobian criterion. This will fit better the intuition that a (smooth) differential manifold is locally an affine space, but that algebraically we do not have the inverse function theorem, hence we "merely" put the jacobian criterion. Give useful examples and mention [Sta23, Tag 00TN]. Mention the fact that a regular finite type schemes over a perfect field are smooth [Sta23, Tag 00TV].

Talk 3. (Oct. 27, Adéla Heroudková) Picard group and divisors topics

Aim : we will recall/define (depending on what people have already seen) Picard group, Cartier divisors, effective Cartier divisors, Weil divisors and the link between them.

As it is a lot to cover, I do not expect all the proofs. Try to give at least a few, but mainly on stating properly the definitions and main results, as well as giving an intuition of how the notions behaves.

We first define the Picard group of a scheme. For that you can look at [LE02, §5.1, Exer. 1.12], [Duca, §3.3.1-3.3.6] which is very well written but in French, or take hints from [Sta23, Tag 01CR] being careful that it is written in the broader setup of ringed spaces. We then define effective Cartier divisors [Sta23, Tag 01WR] [Sta23, Tag 01WS] [Sta23, Tag 01WT]. Give (without proof) the definition of the sheaf of stalks of meromorphic functors $\mathcal{K}_X$ in [LE02, §7.1.1], the definition of Cartier divisors [LE02, §7.1.2, Def. 1.17] and explain how the two definitions of effective Cartier divisors coincide. Reach the definition of Cartier class group and its comparison with Picard group [LE02, §7.1.2, Cor. 1.19] in the reduced case. Follow [LE02, §7.2.1] to define Weil divisors and the class group, then compare everything to the Cartier version [LE02, §7.2.1, Prop. 2.14 and 2.16]. You can skip all the projective examples.

Talk 4. (Nov. 3, Jakob Gerth) What does it mean to blow up... and some genus too

Aim : we will define and study the notions of arithmetic and geometric genus then of blow-ups.

Recall the sheaf of differentials $\Omega^1_{X|S}$ following [Sta23, Tag 00RM] until Lemma 10.131.4 excluded. Mention [LE02, §6.2.1, Prop. 2.5] for further use. Complete the picture by the criterion of smoothness for finite type scheme over a field following [LE02, §6.2.1, Prop. 2.2].

Then define arithmetic of a projective (non necessarily smooth) curve over a field k. For that, investigate [Sta23, Tag 0BY6]. It can be interesting to mention why the cohomology groups are finite dimensional over k (use [Sta23, Tag 02O6] as a projective map is proper) as well as why $H^0(X, \mathcal{O}_X)$ is a finite extension of k (use [Sta23, Tag 0BUG] remembering that a curve is a variety hence integral). Ignore the dualizing sheaf $\omega_{X|k}$ and just state [Sta23, Tag 0C1A] which is an easier computation when X is smooth. Try to run $\mathbb{P}^1_k$.

In a third time, we want to define the blow-up of a closed subscheme and its universal property. Follow [Sta23, Tag 01OF]. I especially want you to state (and maybe prove) [Sta23, Tag 02OS] [Sta23, Tag 0806] and the first part of [Sta23, Tag 02NS].

Note : you need to do a 1h30 sharp talk as the last half hour will be devoted to running out together computations of genus and blow-ups.

Talk 5. (Nov. 10, Priyanka Jain) Birational maps and their description by blow ups

Aim : we will first push further the analysis of quasi-coherent sheaves on $\mathbb{P}^N_{\mathrm{Spec}(A)}$: global generation and vanishing of cohomology after sufficiently large twist. We will then focus on transfer these results to projective schemes over a noetherian affine scheme. We will then use this notions to study birational morphisms, essentially morphisms of schemes over our Dedekind scheme which are model of the same scheme. We will prove that these maps are blow-up maps.

In a first part, you should prove, in the case of $X = \mathbb{P}^N_{\mathrm{Spec}(A)}$, that quasi-coherent sheaves are globally generated after big enough twist. Please try to write down a precise proof mixing the ideas of [LE02, §5.1.4, Lem. 1.25, Th. 1.27] but staying in our specific case for clarity. Then, explain how to deduce the vanishing of cohomology of quasi-coherent sheaves after big enough twist when A is noetherian (follow [LE02, §5.3.1, Th. 3.2] after mentioning [LE02, §5.1.4, Cor. 1.28]).

In a second time, explain why a projective map $X \to \mathrm{Spec}(A)$ must factor by a closed immersion $X \hookrightarrow \mathbb{P}^N_{\mathrm{Spec}(A)}$ (the proof of [Sta23, Tag 087S] works without the ample sheaf). You can hence deduce the results of the first paragraph for a projective map over an affine noetherian scheme (using the reduction in the proof of [LE02, §5.3.1, Th. 3.2]).

We then define (shortly) birational maps and morphisms ([Sta23, Tag 01RN] [Sta23, Tag 01RR]. To help yourself, you can remember that we will can about fibered surface, which are integral and hence can be assumed irreducible. The last task of the talk is to prove [LE02, §8.1.3, Th. 1.24] for X projective over an affine noetherian scheme. For that, please prove [LE02, §8.1.3, Th. 1.24], state [LE02, §8.1.3, Th. 1.24] and prove the first part, and finally prove [LE02, §8.1.3, Th. 1.24]. You should highlight where we used that the map is birational; you should simplify the proof as $\mathcal{M} = i^*\mathcal{O}(d)$ for some closed immersion i of X in a projective space suffices to obtain $\mathcal{J} \otimes \mathcal{M}$ globally generated (it was proven in the first part of the talk). Mention [Sta23, Tag 0C4Q] to deduce the useful result for this seminar : a birational morphism of arithmetic surfaces is a blow-up.

Talk 6. (Nov. 17, Yiqing Yang) Models can be normies

Aim : we finally define properly what we mean by a model of a normal projective curve over the fraction field of a Dedekind scheme of dimension 1*, then explain the existence of normal/regular models for when the curve is smooth and the Dedekind scheme affine.*

First define models [LE02, §10.1.1 Def. 10.1] and give examples illustrating how bad their properties can be. One can be the "unconventional" model of $\mathbb{P}^1_{\mathbb{Q}}$ obtained by blowing up a closed point on the special fiber of $\mathbb{P}^1_{\mathbb{Z}}$ at p that we encountered several times. It produces a regular non-smooth model, when $\mathbb{P}^1_{\mathbb{Z}}$ was a smooth model. If time allows, you can find a non-regular model among [LE02, §10.1.1, Ex. 1.12, Ex. 1.14].

The rest of the talk is devoted to proving that for an affine Dedekind scheme $\mathrm{Spec}(A)$ of dimension 1, whith field of fractions F, any smooth projective connected curve $\mathcal{C}$ over $\mathrm{Spec}(F)$ has a regular model. This goes in four steps :

1. We construct an $\mathrm{Spec}(A)$-scheme C_0, with special fiber $\mathcal{C}$, as a closed subscheme of $\mathbb{P}^N_{\mathrm{Spec}(A)}$. Use Talk 5 to see that $\mathcal{C}$ falls in the setup of [LE02, §10.1.1 Ex. 10.4].
2. Take C_1 to be the closure of the open $\mathcal{C} \subset C_0$, with reduced structure. Prove that it is a fibered surface over $\mathrm{Spec}(A)$, i.e. that it is integral projective flat of dimension 2, with special fiber $\mathcal{C}$. Everything but dimension should be accessible by expliciting what closure does on a affine of C_0. For dimension, you might wanna use [Sta23, Tag 0B2I].
3. For a qcqs map $f : Y \to X$, define the normalisation of X in Y following [Sta23, Tag 0BAK] (you might need [Sta23, Tag 01LC] to understand why [Sta23, Tag 035H] makes sense). Now, you can define C_2 as the normalisation of C_1 in $\mathcal{C}$ and explain why it is a model of $\mathcal{C}$. Showing that C_2 is integral normal and flat should be quite explicite from what happens on affines. To prove that $C_2 \to \mathrm{Spec}(A)$ is projective, restrict to proving that $C_2 \to C_1$ is an use [Sta23, Tag 0AVK] [Sta23, Tag 0B3I]. Same trick that for 2) for the dimension.
4. To move on to regular models, on need the theory of desingularisation. This is a construction that is not that hard to describe, even though the proof of its correctedness is too hard for this seminar. In the talk, we will hence only state the definition [LE02, §8.3.4 Def. 3.39], then explain the construction before [LE02, §8.3.4 Th. 3.44] and state that it abutes. Maybe try to avoid talking about excellent scheme and rather explain why the singular locus is closed for a model of a smooth curve. As desingularisation is a sequence of blow-ups and normalisations, it should not be hard with Talk 4 and step 3) to prove that the obtained C_3 is a normal model of $\mathcal{C}$.

If times allows, find an interesting example and run some explicit computations.

Talk 7. (Nov. 27th, 10 am, Haixu Wang) Intersection theory on surfaces

Aim : we study intersection theory on surfaces, giving a combinatorial data about how two divisors in a surface intersect. This seminar will further use it to derive combinatorial data from the special fibers of models.

One first goal is to prove the refinement of Talk 5's result about birational maps [LE02, §9.2.1, Th. 2.2]. Try to orient the proof towards understand how the various divisors come into play, using [LE02, §8.3] as a blackbox. It is both a very useful result and a good motivation about wether the behavior of Cartier divisors, especially vertical ones, is crucial.

Recall how we defined "easily" the multiplicity of a Cartier divisor at a point of codimension 1 on a regular locally Noetherian scheme in Talk 3. A first step of this talk is to give a broader definition of mult_x following the end of [LE02, §7.1.2]. Take a shortcut by only explaining the first statement of [LE02, §7.1.1, Lem. 1.2] and the second of statement of [LE02, §7.1.1, Cor. 1.3]. Say why it recovers our previous definition.

Building on that define the intersection number of two effective Cartier divisor at a closed point of a noetherian connected scheme X of dimension 2, following [LE02, §9.1.1]. Be especially careful to unravel

why the first paragraph of *loc. cit.* is true. Give examples to strengthen everybody's intuition. Prove [**f**]or a X regular. It is especially interesting that b) links the intersection number to some properties happening on the divisor E.

In a second time, recall the degree of a Cartier divisor on a compact curve over a field k in [LE02, §7.3.1, Def. 3.1]. Be careful to justify why it is correctly define, maybe by using the arguments at the end of [LE02, §7.1.2]. State, without proving that these is a notion of degree for line bundles that recover the degree of Cartier divisors [LE02, §7.3.2, Lem. 3.30]. Run the example of $\mathbb{P}^1_k$ quickly. For a closed point s of a Dedekind scheme S of dimension 1. Using that, we can define intersection number of a vertical divisor and another divisor for a regular arithmetic surface X : the condition [LE02, §9.1.2, Th. 1.12 (d)] hence makes sense and there is at most one bilinear map $\mathrm{Div}(X) \times \mathrm{Div}_s(X)$ extending it. Prove the rest of [LE02, §9.1.2, Th. 1.12]. Roughly, this allows to define intersection number also for divisors having some common irreducible components, in particular the self-intersection number of vertical divisors in [LE02, §9.1.2, Def. 1.15]. An easy consequence is [LE02, §9.1.2, Prop. 1.21].

Talk 8. (Dec. 1, Gun Kongarttagarn) These divisors are really exceptionnal or how to reach minimal models

Aim : explain how to obtain minimal regular models (roughly every regular models have a birational morphism to them) by contracting a specific type of curve.

Define the notion of contraction of a family of integral vertical curves [LE02, §8.3.3 Def. 3.27]. We aim to reach the criterion [LE02, §8.3.3 Prop. 3.30 (ii)⇒ (i)] and proving it quite carefully, which need to go up accordingly in the loc. cit. book.

We then brush [LE02, §9.3.1] defining the notion of exceptionnal divisors, i.e. divisors that we can contract (hence proving that the surface is not minimal). The paragraph explain that it is always the blow-up at a point, using results of Talk 5 and compute everything in this case, showing that $E \simeq \mathbb{P}^1_{k'}$ and $E^2 < 0$. It shows that we can always contract such divisor but that regularity is not necessarily preserved, ending up with a characterisation of such exceptionnal divisors called Castelnuovo's criterion [LE02, §9.3.1 Th. 3.8].

We then investigate the notion of relatively minimal surfaces by brushing [LE02, §9.3.2], analysing how to obtain representatives of birational morphisms classes [LE02, §9.3.2 Prop. 3.19] by successively contracting exceptionnal divisors.

Finally, we brush [LE02, §9.3.3] which covers the notion of minimal model. It boils down to proving that two relatively minimal models are isomorphic [LE02, §9.3.3 Th. 3.21].

As the three last paragraphs represent a lot to cover, we do not expect to make all the proof. Try to quote all the results mentionned, and to prove in details one of the paragraphs at least.

Talk 9. (Dec. 8, Tomáš Perutka) Semistable models under scrutiny

Aim : we define semistable algebraic varities over $\mathbb{F}_p$, which are roughly the one étale locally looking like crosses $\mathrm{Spec}(k[x,y]/xy)$. *We also study semistable models over $\mathbb{Z}_p$, which are roughly étale locally looking like annuli of integer length.*

The first part of the talk is to define nodes, also called singular ordinary double points of a variety over an algebraically closed field. For this, we can mix [LE02, §7.5] (from the paragraph before Lem. 5.12 to Prop. 5.15) and [Sta23, Tag OC1P] and focus on double points. We will make appear the equivalence between the description by δ and branches invariants and the description of the stalks.

Then, we define semistable curves over a field, also called at-worst-nodal. One can compare [LE02, §10.3.1 Def. 3.2] and [Sta23, Tag 0C47]. Run out [LE02, §10.3.1 Ex. 3.4]. We can mention [LE02, §10.3.1 Prop. 3.7] if we have time. Finally, we reach the definition of a smooth projective curve having a semistable model, equivlently having at-worst-nodal families of curves as model. One can once again compare the compact are

clear [Sta23, Tag 0CDB] [Sta23, Tag 0C58] and [LE02, §10.3.2] which uses less frightening terms. There is no need to cover everything in this paragraph, but it is important to highlight the étale local structure of at-worst-nodal families of curves [LE02, §10.3.2 Cor. 3.22] and the link with minimal models [LE02, §10.3.2 Th. 3.34] which both have analogues in Stacks Project.

Talk 10. (Dec. 15, William Hendareo) Proof of weak Deligne-Mumford part I : combinatorial invariants

Aim : we associate combinatorial data to the special fiber of an arithmetic surface, define two combinatorial invariants β and a "new" genus, and make the link with the arithmetic genus and the picard group.

Begin by stating a weak Deligne-Mumford theorem that will be the aim of the next two talks. It is [LE02, §10.4 Th. 4.3] for S being the spectrum of a discrete valuation ring and semistable instead of stable.

First, we dig into [LE02, §10.1.4] aiming to understand [LE02, §10.1.4 Ex. 4.56]. We advice to present things in a reverse order. First explain what data are given in [LE02, §10.1.4 Ex. 4.56] and their properties that will suit the definition of a type [LE02, §10.1.4 Def. 4.55]. For that, you can help yourself with [Sta23, Tag 0C5Y] (while being careful as the definition of type in Stacks is more involved). Then, one can define the type of (the special fiber) of an arithmetic surface [LE02, §10.1.4 Def. 4.55] and its genus. We finish this first part by constructing the graph associated with a type as well as the invariant β. This can be made by following or selecting among [LE02, §10.1.4] up to [LE02, §10.1.4 Prop. 4.51]. The first two items of this property should be explained.

In a second time, we investigate the link between the Picard group of an arithmetic surface X over a discrete valuation ring, its special fiber and invariants that has to do with the type. This is [LE02, §10.4.2] until [LE02, §10.4.2 Prop. 4.17].

Talk 11. (Dec. 22, Ernesto Thimm) Proof of weak Deligne-Mumford part II ;

Aim : we finish the proof of the weak Deligne-Mumford theorem by studying abelian and toric rank of a regular model.

We define the abelian and toric rank of an arithmetic surface by covering [LE02, §10.4.2 5.17-5.24]. Remember to restrict to arithmetic surface to not state/prove everything with the greatest generality and details.

Taking [LE02, §10.4.2 Th. 4.19] as a blackbox, we explain how to relate the toric rank and the construction of last talk in [LE02, §10.4.2 Prop. 4.20].

Finish the proof as in [LE02, §10.4.2 p.542].

If time allows, we can prove the genus 0 case using [Sta23, Tag 0CDK]. If time still allows, we can brush the genus 1 case extracting the core ideas from [Sta23, Tag 0CEG].

Talk 12. (Jan. 12, Gautham Mahesh) Formal schemes

From here, the seminar takes a turn : the upcoming talk are more cultural than formal (only the schemes are). It is useful to make precise definitions when possible, but it is also allowed to drop some technical adjectives in the presentation, possibly mentionning that your dropped them. I rather expect you to focus

on why the objects were introduced, to compare them to the notions you know, and to give an intuition about how to think about them.

This talk first aims to give an idea about how formal geometry serves as the correct replacement for model over $\mathrm{Spec}(\mathbb{Z}_p)$ and how it can give rise to rigid geometry over $\mathbb{Q}_p$. Trying to focus on the case of p-adic formal schemes, we can freely get inspiration from [FK18, Intro §0, 3-6] and [Ill03, §1]. One should definitely highlight how we keep only the topological information of the special fiber while still keeping $\mathbb{Z}_p$-algebras (usually non torsion) as structure sheaves. Then, we will try to give a broad overview of [FK18, p. I 1.1, 1.2, 1.4, II 1.1], and give some (but not too much) clean definition. Finish by an analysis of the canonical formal scheme sent by localisation to the unit ball, i.e. the p-adic formal scheme

$$\mathrm{Spf}(\mathbb{Z}_p\langle t\rangle), \text{ where } \mathbb{Z}_p\langle t\rangle := \left\{ \sum_{n\geq 0} c_n t^n \;\middle|\; c\in\mathbb{Z}_p^{\mathbb{N}},\ c_n \xrightarrow[n\to+\infty]{} 0 \right\}.$$

Talk 13. (Jan. 19, Angwei Li) Rigid analytic spaces from Berkovich's point of view

From here, the seminar takes a turn : the upcoming talk are more cultural than formal. It is useful to make precise definitions when possible, but it is also allowed to drop some technical adjectives in the presentation, possibly mentionning that your dropped them. I rather expect you to focus on why the objects were introduced, to compare them to the notions you know, and to give an intuition about how to think about them.

This talk first aims to give the main features of Berkovich theory of rigid analytic spaces over $\mathbb{Q}_p$ and how it relates to the previous approach. To motivate, we can freely get inspiration from [Duc06, Intro and §1] and [FK18, Intro §7]. We advice to run a precise definition of a $\mathbb{Q}_p$-analytic Berkovich space but to keep other definition vague. In terms of reading and presenting, I advice to follow a precise path. First, read until [Duc06, Def. 1.4] to reach the definition of the topological space of an affinoid Berkovich space. We should then run out an explicit description of $\mathbb{A}^{1,\mathrm{an}}_{\mathbb{Q}_p}$ as in [Poi16]. We can compare the classification of the points to the (more technical) one given in [Ducb, §3.3], if time allows. Give other examples without the full description like the polydisks [Duc06, Ex. 1.6], circles [Duc06, Ex. 1.3], the annuli [Duc06, Ex. 1.14]. We can then brush the actual construction of $\mathbb{Q}_p$-analytic Berkovich spaces and their structure sheaf by following [Duc06, §1.3 until Def. 1.24 excluded]. Finally, try to explain the generic (analytic) fiber of a finite type p-adic formal scheme as in [Duc06, §1.4].

Note that you can also pick among Berkovich original articles but they are less beginner friendly.

Note : the talk should be 1h 30 sharp as the last hour will be devotes to some computations together.

Talk 14. (Jan. 26, Nataniel Marquis) A pencil, drawing triangulations and you got your semistable model

You will discover the details in due time.

Références

[Duca] Antoine Ducros. *Introduction à la théorie des schémas*. Lecture notes available at webusers.imj-prg.fr/ antoine.ducros.

[Ducb] Antoine Ducros. *La structure des courbes analytiques*. draft of a book in preparation. URL : http://webusers.imj-prg.fr/~antoine.ducros/trirss.pdf.

[Duc06] Antoine Ducros. *Espaces Analytiques p-adiques au sens de Berkovich*. exposé 958 du séminaire Bourbaki. 2006. URL : https://webusers.imj-prg.fr/~antoine.ducros/asterisque.pdf.

[FK18] Kazuhiro Fujiwara et Fumiharu Kato. *Foundations of Rigid Geometry I*. European Math. Soc., 2018.

[Ill03] Luc Illusie. *Grothendieck's existence theorem in formal geometry*. online notes for Advanced School in Basic Algebraci Geometry. 2003. URL : https://indico.ictp.it/event/a0255/session/14/contribution/9/material/0.

[LE02] Qing Liu et Reinie Erné. *Algebraic Geometry and Arithmetic Curves*. Oxford University Press, mai 2002. DOI : 10.1093/oso/9780198502845.001.0001.

[Mat80] Hideyuki Matsumura. *Commutative Algebra*. Benjamin/Cummings Publishing Company, 1980.

[Poi16] Jérôme Poineau. "Raconte-moi... la droite de Berkovich". In : *Gazette de la S.M.F.* (2016). URL : https://smf.emath.fr/sites/default/files/2020-05/Poineau-site.pdf.

[Sta23] The Stacks project authors. *The Stacks project*. https://stacks.math.columbia.edu. 2023.